

Modern R for Public Health Practitioners

CT DPH DMAG Training

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Table of contents

Welcome	3
Special Thanks	3
About the Author	3
Preface	4
I Introduction and Process	5
1 Adopting an R Project Mindset	6
1.1 The Mental Model	6
2 How to use this book	7
2.1 The big idea	7
2.2 The arc	7
2.3 How to use this site	8
2.4 Before you start	8
2.5 What you'll need	8
II Working with real data	9
3 Getting Ready for working with files	10
3.1 The Setup	10
3.2 Data Intro	10
3.3 Setup	10
3.3.1 Installing Packages	11
4 Reading Comma Delimited Files	12
4.1 Reading our Comma Delimited File (.csv)	12
Appendices	14
A Resources	14
B Tools	15

Welcome

This is the website for the Connecticut Department of Public Health's Data Management, Analytics, and Governance (DMAG) R training for public health practitioners.

Special Thanks

- Gary Archambault and the rest of the DMAG team
- Mukhtar Mohamed
- Sydney Jones

About the Author

Epidemiologist turned Data Scientist with a decade of R experience.

Preface

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Part I

Introduction and Process

1 Adopting an R Project Mindset

1.1 The Mental Model

2 How to use this book

This course is not about teaching you epidemiology. It's about giving you the R fluency to apply what you already know.

2.1 The big idea

We will dive straight into a data analysis project and with every chapter will we iteratively polish our analysis until at the very end we have something that could be confidently deployed in day to day use.

2.2 The arc

WIP - PLACEHOLDER

Chapter	What you build	The new skill
0 — Orientation	Your first rendered R analysis	R mindset; the SAS→R Rosetta stone
1 — Import	A real CT dataset + data profile	<code>readr</code> ; inspect before you trust
2 — Wrangling	A clean, joined, analysis-ready dataset	tidyverse DATA -step verbs; joins
3 — Descriptive epi	Publication-quality tables	<code>gt</code> ; counts, rates, CIs
4 — Visualization	Trend charts + a CT choropleth map	<code>ggplot2</code> ; <code>sf</code> + <code>tigris</code>
5 — Quarto report	Leadership-ready report v1	Quarto; findings before methods
6 — Functions	One template, any county or year	Functions; Quarto params
7 — API	A self-fetching, live-data report	SODA API; SoQL; fallback caching
8 — Reproducibility	Published GitHub Pages capstone	<code>renv</code> ; <code>here</code> ; GitHub Pages
9 — Automation <i>(stretch)</i>	Scheduled re-render	GitHub Actions; handoff

2.3 How to use this site

Work through modules in order. Each one has:

- **Overview** — what you’ll build, why it matters, and the SAS concepts it remaps
- **Lesson** — a runnable, narrated `.qmd` file you execute yourself
- **Exercises** — 3–5 scaffolded problems with explicit “done when” checks
- **Solutions** — available after you’ve tried on your own

2.4 Before you start

 Start here:

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2.5 What you’ll need

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Part II

Working with real data

3 Getting Ready for working with files

3.1 The Setup

Covid-19 challenged Public Health infrastructure in so many multi-faceted ways. The sheer amount of data collected and processed over the pandemic meant that what worked in the beginning of the pandemic was no longer feasible towards the end of the pandemic.

Over the course of this book we will explore Connecticut pandemic data and how to deal with the ever increasing amount of data. We will start simple and do things manually before moving on to refine our analytics methodologies.

3.2 Data Intro

We are going to use a snippet of real pandemic-era covid-19 data from the connecticut department of public health's open data portal¹.

The full dataset can be found on the portal² but we will start simply and use the provided snippet in this

TODO: add evergreen link

repository `chapters/working-with-real-data/data/covid-2022.csv`

3.3 Setup

R has many tools for working with data. Over the years the R community has built up a number of their own tools and one of the most popular, the `tidyverse` will be our entrypoint into working with data in R.

R comes with a base set of tools and eager epidemiologists can opt to use only the base suite of tool and go out into the world if they so choose. However, many smart developers and epidemiologists have come before us and we do not need to recreate the wheel. R has what are called 'packages' - or collection of ready made code that folks can use to get work done.

¹[CT Open Data Portal](#)

²[Archived Covid Data](#)

TODO: – talk about the CRAN etc etc....

We're going to slowly introduce the different parts of the tidyverse individually....

TODO: flesh

3.3.1 Installing Packages

```
# pulls the code down from the CRAN for folks to use and installs on
# their computer
install.packages(
  "dplyr", # for manipulating data
  "readr", # for reading data files from a users' computers
  "fs" # for working with files on users' file systems
)
```

4 Reading Comma Delimited Files

4.1 Reading our Comma Delimited File (.csv)

We are going to do a few things here:

1. 'Invoke' the packages we will be using so they are available to us.
2. Create and capture the path to our data file and save inside of an object `csv_path`.
3. Read the data into our session from the csv and save inside of an object `covid_data`.

```
# Users can have all sorts of packages on their computer
# but will only need a limited subset of them at any one time
# add these packages to your session with `library()` function.

library(dplyr)
library(readr)
library(fs)

# we have a file - how do we read it in?
# we use fs's path() function to create a path
# we use readr's read_csv() and point it to our path

csv_path <- path("data/covid-2022.csv")

covid_data <- read_csv(csv_path)
```

We have our data saved into our session, now what? Let's take a look at it!

```
# taking a look at the top few rows
head(covid_data)
```

```
# A tibble: 6 x 16
  last_update_date town_number town    total_cases confirmed_cases probable_cases
  <date>           <dbl> <chr>         <dbl>         <dbl>         <dbl>
1 2022-01-03         1 Andov~         312           269           43
```

2	2022-01-03	2 Anson~	3103	2632	471
3	2022-01-03	3 Ashfo~	441	398	43
4	2022-01-03	4 Avon	1690	1584	106
5	2022-01-03	5 Barkh~	325	309	16
6	2022-01-03	6 Beaco~	888	790	98

```
# i 10 more variables: case_rate <dbl>, total_deaths <dbl>,
#   confirmed_deaths <dbl>, probable_deaths <dbl>, people_tested <dbl>,
#   rate_tested_per_100k <dbl>, number_of_tests <dbl>,
#   number_of_positives <dbl>, number_of_negatives <dbl>,
#   number_of_indeterminates <dbl>
```

```
# glimpse from dplyr
glimpse(covid_data)
```

Rows: 20,956

Columns: 16

```
$ last_update_date      <date> 2022-01-03, 2022-01-03, 2022-01-03, 2022-
01-~
$ town_number           <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14~
$ town                  <chr> "Andover", "Ansonia", "Ashford", "Avon", "Bar~
$ total_cases            <dbl> 312, 3103, 441, 1690, 325, 888, 2580, 646, 27~
$ confirmed_cases       <dbl> 269, 2632, 398, 1584, 309, 790, 2392, 584, 22~
$ probable_cases        <dbl> 43, 471, 43, 106, 16, 98, 188, 62, 494, 71, 2~
$ case_rate              <dbl> 9642, 16635, 10364, 9247, 9013, 14272, 12625,~
$ total_deaths           <dbl> 4, 35, 5, 76, 7, 7, 44, 4, 63, 2, 124, 6, 7, ~
$ confirmed_deaths       <dbl> 3, 31, 4, 61, 6, 7, 38, 3, 49, 2, 104, 4, 7, ~
$ probable_deaths        <dbl> 1, 4, 1, 15, 1, 0, 6, 1, 14, 0, 20, 2, 0, 7, ~
$ people_tested          <dbl> 2243, 14847, 3105, 15488, 2487, 4580, 15122, ~
$ rate_tested_per_100k   <dbl> 69314, 79592, 72973, 84745, 68968, 73610, 739~
$ number_of_tests        <dbl> 7638, 62140, 10619, 59793, 9080, 19193, 62374~
$ number_of_positives     <dbl> 352, 3735, 510, 1959, 379, 1059, 3095, 779, 3~
$ number_of_negatives     <dbl> 7281, 58269, 10099, 57770, 8689, 18094, 59221~
$ number_of_indeterminates <dbl> 5, 136, 10, 64, 12, 40, 58, 20, 59, 16, 162, ~
```

A Resources

B Tools